Arduino-Based Power Measurement System

Version 1 Prototype — Code and Circuit Documentation

1. Introduction

Version 1 is a prototype for measuring the electrical quantities of a load using an Arduino Uno R3 and an INA219 current/voltage sensor. The sensor communicates with the Arduino over the I²C bus and reports bus voltage, shunt voltage, current, and power. In the illustrated circuit, an LED and a 220 Ω resistor form the load path monitored by the sensor. The Arduino outputs the measured values to the Serial Monitor for observation.

2. Circuit Configuration

The Arduino supplies 5 V and ground to the INA219 module. The I²C data line (SDA) connects to Arduino A4, while the clock line (SCL) connects to A5. The load current path runs from the 5 V supply to VIN+, through the INA219's current-sense path, and out of VIN− to the 220 Ω resistor, LED, and ground. The sensor therefore measures the load current in series with the LED circuit. The Arduino ground and load ground share a common reference.

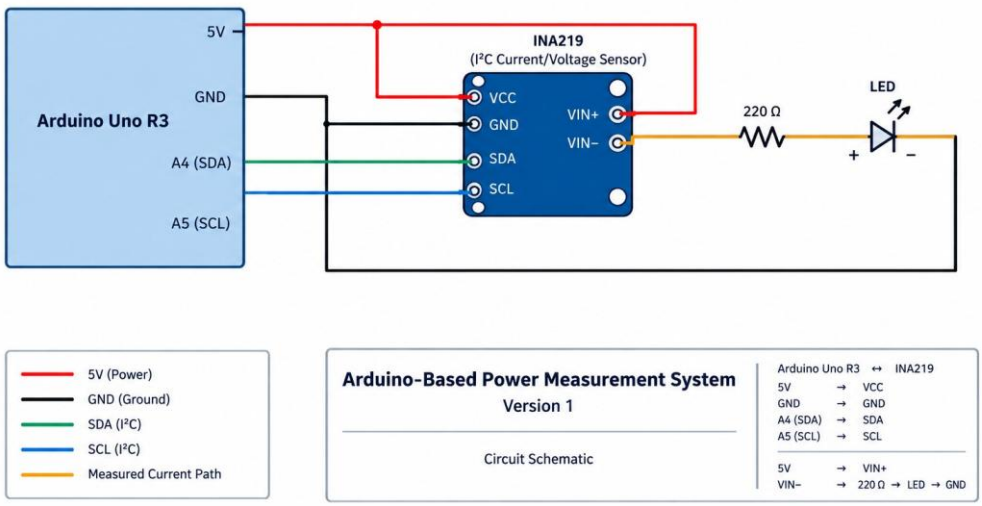


Figure 1. Version 1 circuit schematic.

3. Program Operation

3.1 Libraries and sensor object

The Wire library provides I²C communication support. The Adafruit_INA219 library provides the interface used to initialise the sensor and retrieve its measurements. The declaration `Adafruit_INA219 ina;` creates the sensor object referenced throughout the program.

3.2 Initialisation in setup()

The program starts serial communication at 9600 baud and prints a startup message. It then calls `ina.begin()` to initialise the INA219. If the sensor is not detected, an error message is printed and the program remains in an infinite loop, preventing normal measurement operation. If initialisation succeeds, a confirmation message is sent to the Serial Monitor.

3.3 Measurement loop

Every loop iteration reads four values from the sensor: bus voltage in volts, shunt voltage in millivolts, current in milliamperes, and power in milliwatts. Each value is printed with a descriptive label and its corresponding unit. The `delay(5000)` call pauses the program for 5,000 milliseconds, so the readings are normally updated at intervals of approximately five seconds, excluding execution time.

4. Measurement Quantities

Bus voltage: The voltage measured at the bus input of the INA219, reported in volts (V).
Shunt voltage: The small voltage drop across the sensor's current-sense resistor, reported in millivolts (mV).
Current: The load current calculated by the INA219 from the shunt measurement and its configuration, reported in milliamperes (mA).
Power: The power value reported by the INA219 library, reported in milliwatts (mW).

Voltage, current, and power use different units in the serial output. These units are part of the measurement labels and should be retained when recording or interpreting results.

5. Annotated Code

```
#include <Wire.h>
#include <Adafruit_INA219.h>

Adafruit_INA219 ina;

void setup() {   Serial.begin(9600);
Serial.print("\nINA219 Starting....");

  if (!ina.begin()) {
Serial.print("\nERROR: INA219 not found");
while (1) {       delay(10);     }
  }

  Serial.print("\nINA219 connected\n");
}

void loop() {   float shuntV =
ina.getShuntVoltage_mV();   float BusV =
ina.getBusVoltage_V();   float Current =
ina.getCurrent_mA();   float Power =
ina.getPower_mW();

  Serial.print("\nBus Voltage: ");
  Serial.print(BusV);
  Serial.println(" V");

  Serial.print("Shunt Voltage: ");
  Serial.print(shuntV);
  Serial.println(" mV");

  Serial.print("Current: ");
  Serial.print(Current);
  Serial.println(" mA");

  Serial.print("Power: ");
  Serial.print(Power);
Serial.println(" mW");

  delay(5000);
ina.begin(); }
```

6. Output Format

The Serial Monitor displays four labelled measurements in sequence. A typical output block has the following structure; the numerical values depend on the connected load and supply. For example:

```
Bus Voltage: 4.21 V
Shunt Voltage: 0.21 mV
Current: 2.10 mA
Power: 10.00 mW
```

7. Scope of Version 1

The current prototype demonstrates sensor initialisation, acquisition of voltage/current/power measurements, and serial reporting for a single LED load. The documented implementation uses the INA219 library's measurement functions and a fixed five-second delay between output cycles. The final `ina.begin()` call in `loop()` repeats sensor initialisation after each reporting cycle; the initialisation check itself is performed in `setup()`.

8. Pictures

